

The empirical recurrence for OEIS A251332: recovering a conjecture that is not written down

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Abstract

OEIS A251332 records “Empirical recurrence of order 77”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 729$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 77, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 77 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A251332 is “Number of $(n+1) \times (5+1)$ 0..2 arrays with every 2×2 subblock summing to 2, 4 or 6.”. It has offset 1 and begins

14823, 343299, 8863633, 256067697, 8130801167, 277153898751, 9919268234529, 366492859729101, ...

The entry states:

Empirical recurrence of order 77 (see link above).

As of the “Last modified” line on the live entry (Apr 10 2022 15:15:33, revision 6) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 729$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 729$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1458$ exact terms of the model returns order 77 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{77} c_i a(n-i),$$

c_1	275	c_{21}	19582455866490607670391052	c_{41}	−14136576882171366548
c_2	−34345	c_{22}	−104184297754863592692102503	c_{42}	371167177083069021159
c_3	2565781	c_{23}	−604803641655809194720528475	c_{43}	379351353938402859875
c_4	−126724432	c_{24}	2051317657345021199760224159	c_{44}	−58407303628883935977
c_5	4292522395	c_{25}	14685899404014332688019801641	c_{45}	−70651618803505213563
c_6	−98782830473	c_{26}	−21473601305560667312642838199	c_{46}	741992310000946283905
c_7	1413517404314	c_{27}	−254209984101641247688185667199	c_{47}	974429198998186899430
c_8	−7612912632320	c_{28}	1069600625746274705862692070	c_{48}	−78562477813128728355
c_9	−133383749714916	c_{29}	2979085060420242613091956057303	c_{49}	−102197657713469128164
c_{10}	3020499366138663	c_{30}	3209327698074998064598603685128	c_{50}	709534261435170462273
c_{11}	−15498346546695193	c_{31}	−23050694121290049944116212880016	c_{51}	820991680492488977670
c_{12}	−215961707858725792	c_{32}	−46959641845109045716859036004609	c_{52}	−54827155813695693326
c_{13}	3182435178964201363	c_{33}	113600352340335502033665703972821	c_{53}	−50084443438251615988
c_{14}	742224312769079570	c_{34}	371384338955202307553267724784813	c_{54}	355233345870608786884
c_{15}	−255649972072533064746	c_{35}	−313174601085497609135744039898035	c_{55}	225411510639325860562
c_{16}	865959431482882404756	c_{36}	−1916863787738892370194139649642349	c_{56}	−18632115574112489006
c_{17}	13384382789346910149074	c_{37}	105832592488761284751853747838103	c_{57}	−69763760527560434890
c_{18}	−72246147667899162384438	c_{38}	6943909201757866800937448756980054	c_{58}	75949254600228670707
c_{19}	−549069127657465866397440	c_{39}	2972593167321852884063800617850059	c_{59}	118396545927132235714
c_{20}	3383738157632401895724599	c_{40}	−18440034364001964908591684527354177	c_{60}	−23000010483598072981

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 77, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $77 + 4$ terms removed returns order 77 as well, so $\deg q_{\min} = 77 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $77 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 77 that this sequence satisfies — and that family turns out to have exactly one member.

References

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